ModelIt!

Teacher's Guide

Why Do Roller Coasters Go So Fast?

The Amazing Science of Energy and Speed

Quick Reference

Grade Level	Grade 4
NGSS Standard	4-PS3-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: 4-PS3-1

Use evidence to construct an explanation relating the speed of an object to the energy of that object.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how hill height, speed, and energy of motion are connected in a roller coaster system.
Disciplinary Core Idea	**PS3.A Definitions of Energy** — The faster a given object is moving, the more energy it possesses. Energy can be moved from place to place by moving objects or through sound, light, or electric currents.
Crosscutting Concept	**Energy and Matter** — Students explore how energy transforms from potential to kinetic as a roller coaster moves, demonstrating that energy is conserved but changes form.

CAST Testing Alignment

MULTIPLE CHOICE:

A marble rolls down a tall ramp and then a short ramp. It goes faster at the bottom of the tall ramp. Which best explains why?

CONSTRUCTED RESPONSE:

Using what you know about energy, explain why a roller coaster goes fastest at the bottom of the tallest hill on the track. Use the words 'potential energy' and 'kinetic energy' in your answer.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Energy is one of the most important concepts in all of science. Understanding how energy changes form helps us explain everything from roller coasters to rocket ships to why our food gives us the power to run and play.

What Is Energy?

Energy is the ability to cause change or do work. It comes in many forms: kinetic energy (motion), potential energy (stored), thermal energy (heat), sound energy, light energy, and more. Energy is never created or destroyed — it only changes from one form to another. This is called the Law of Conservation of Energy.

Potential Energy and Height

When an object is lifted up high, it gains gravitational potential energy — energy stored because of its position. The higher the object, the more potential energy it has. A roller coaster at the top of a 100-foot hill has much more stored energy than one at the top of a 20-foot hill. This stored energy is like a battery waiting to be used.

Kinetic Energy and Speed

Kinetic energy is the energy of motion. Any moving object has kinetic energy, and the faster it moves, the more kinetic energy it has. When a roller coaster rolls downhill, its potential energy converts to kinetic energy. At the very bottom of the hill, nearly all the stored energy has become motion energy — that is when the coaster is going fastest.

Energy in Everyday Life

Energy transformations happen all around us every day. When you eat food, chemical energy transforms into kinetic energy so you can move. When you turn on a light, electrical energy transforms into light and heat energy. A bouncing ball converts kinetic energy to potential energy and back with every bounce. Understanding energy helps engineers design everything from cars to buildings to playground equipment.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify hill height as an external component and speed at bottom and energy of motion as internal components that change in response to hill height.

E **Establish Relationships**
Students determine that hill height positively affects speed at bottom, and speed at bottom positively affects energy of motion — creating a direct chain of cause and effect.

V **Visualize & Model**
Students build a model showing how increasing hill height leads to greater speed and more energy of motion at the bottom of the hill.

E **Evaluate Outcomes**
Students run small hill, medium hill, and giant hill scenarios to observe how changing one input (height) affects the entire system.

R **Revise & Extend**
Students add friction or coaster mass to explore why real roller coasters eventually slow down even though energy is conserved.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Who here has been on a roller coaster? What is the most exciting part — going up or zooming down?"*

Watch For

Show a short roller coaster video clip. Let students share their experiences briefly.

Slide 2: Learning Objectives

3 min

Say: *"Today we are going to figure out the SCIENCE behind why roller coasters go so fast!"*

Watch For

Have students read objectives aloud. Pre-teach 'kinetic energy' and 'potential energy' with simple examples.

Slide 3: Big Question

3 min

Say: *"Here is the mystery: a roller coaster has NO engine pushing it downhill. So where does all that speed come from?"*

Watch For

Have students turn and talk with a partner: What do you think makes a roller coaster go fast at the bottom?

Slide 4: LEVER Framework

2 min

Say: *"We are going to build a model to crack this mystery — step by step."*

Watch For

Briefly preview each LEVER step. Remind students what a model is and why scientists use them.

Slide 5: Activity 1: Components

8 min

Say: *"What things affect how fast a roller coaster goes? Let us sort them into things we can control and things that happen inside the system."*

Watch For

Guide sorting of three components. Explain that hill height is external because WE decide how tall to build the hill.

Slide 6: Activity 2: Connections

8 min

Say: *"When the hill gets taller, what happens to the speed? Does speed go UP or DOWN?"*

Watch For

Help students draw positive arrows from hill height to speed, and from speed to energy of motion.

Slide 7: Activity 3: Simulation

10 min

Say: *"Time to test our model! Let us start with a tiny hill and work our way up to the biggest hill possible."*

Watch For

Have students predict first, then run each scenario. Compare small vs. medium vs. giant hill results.

Slide 8: Discoveries

5 min

Say: *"So NOW we know the secret — it is all about ENERGY changing form!"*

Watch For

Summarize findings. Use a ball and ramp to physically demonstrate energy conversion.

Slide 9: STEM Challenge

12 min

Say: *"Now YOU are the engineers! Design the fastest marble run you can."*

Watch For

Distribute materials (cardboard tubes, tape, marbles). Groups design and test marble runs using model evidence.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Hill Height
- **INTERNAL:** Speed at Bottom, Energy of Motion

Relationships

- **Hill Height to Speed at Bottom:** POSITIVE (+) — A taller hill gives the coaster more stored potential energy. As it rolls down, more stored energy converts to more speed. Double the height means noticeably more speed at the bottom.
- **Speed at Bottom to Energy of Motion:** POSITIVE (+) — The faster the coaster moves, the more kinetic energy it has. Speed and energy of motion always increase together — you cannot have high speed without high energy.

Key Discoveries

- Taller hills give the roller coaster more stored energy at the top, which turns into more speed at the bottom
- Speed and energy of motion are connected — when speed goes up, energy of motion goes up too
- Energy is never created or destroyed — it just changes from one form to another
- A roller coaster at the top of a hill has lots of potential energy that converts to kinetic energy as it rolls down

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Roller coasters have engines that push them the whole time"	Most roller coasters only use a motor for the first hill (the chain lift). After that, gravity and energy do all the work! The coaster converts stored potential energy into kinetic energy as it goes downhill.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Heavier things always go faster"	On a frictionless ramp, all objects fall at the same rate regardless of weight — Galileo proved this! In real life, heavier objects may roll slightly differently because of friction, but height is what really determines speed at the bottom.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Energy gets used up and disappears"	Energy is NEVER destroyed — it only changes form. When a roller coaster slows down, the kinetic energy does not vanish. It transforms into heat energy from friction and sound energy from the rumbling wheels. The total amount of energy stays the same.